Gattoc, Mika Mclaren O.

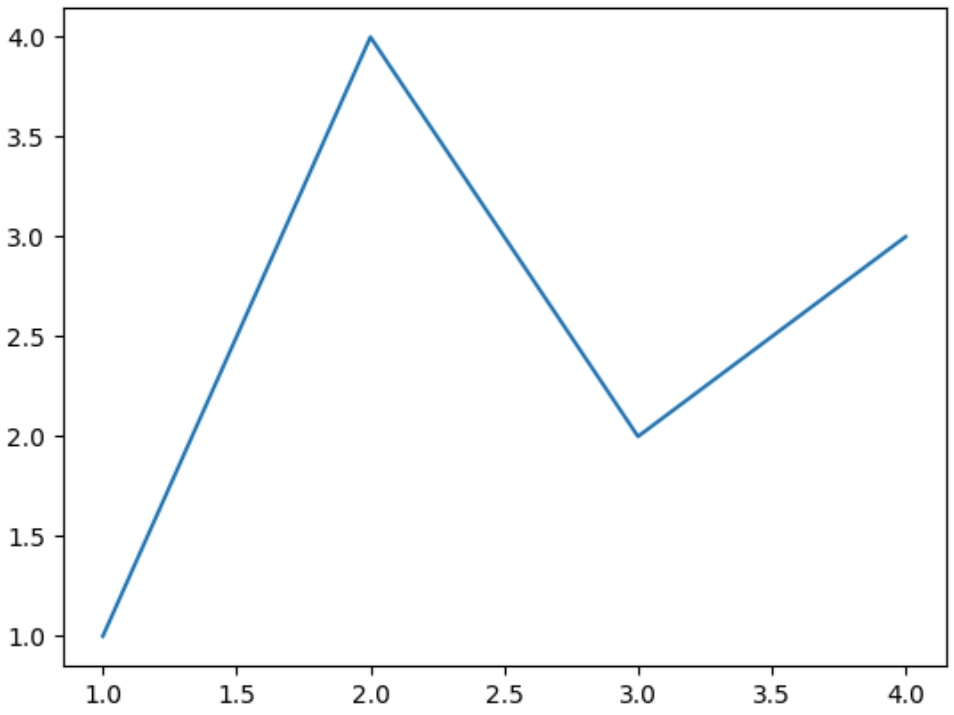
ITCC503-303i

ITE Elective 2

Matplotlib Documentation

```
import matplotlib.pyplot as plt

fig, ax = plt.subplots()
ax.plot([1, 2, 3, 4], [1, 4, 2, 3])
plt.show()
```

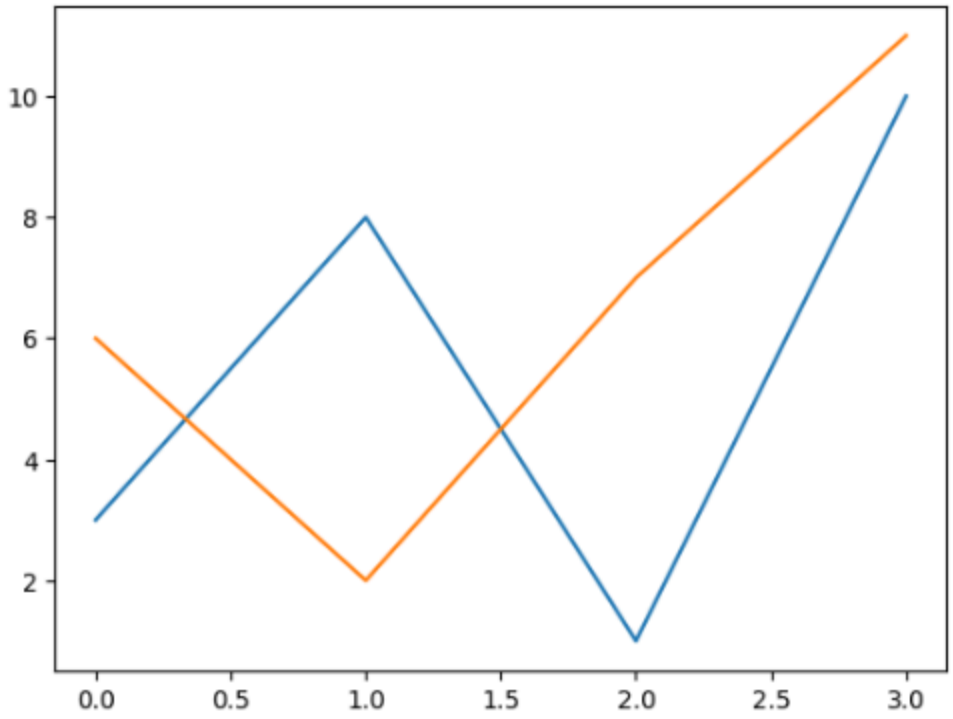


```python
import matplotlib.pyplot as plt
import numpy as np

y1 = np.array([3, 8, 1, 10])
y2 = np.array([6, 2, 7, 11])

plt.plot(y1)
plt.plot(y2)

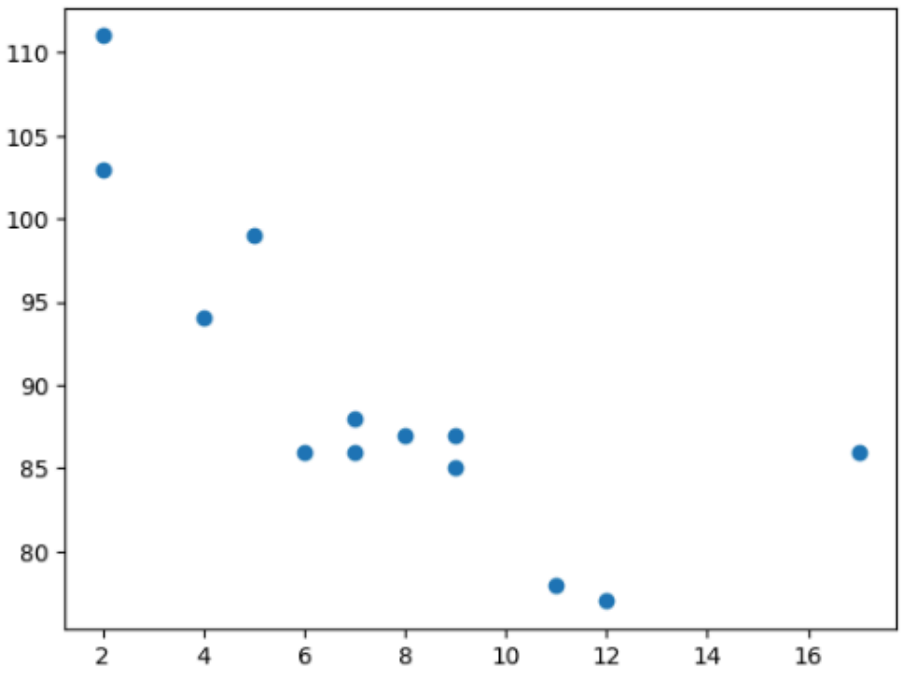
plt.show()
```



```python
import matplotlib.pyplot as plt
import numpy as np

x = np.array([5, 7, 8, 7, 2, 17, 2, 9, 4, 11, 12, 9, 6])
y = np.array([99, 86, 87, 88, 111, 86, 103, 87, 94, 78, 77, 85, 86])

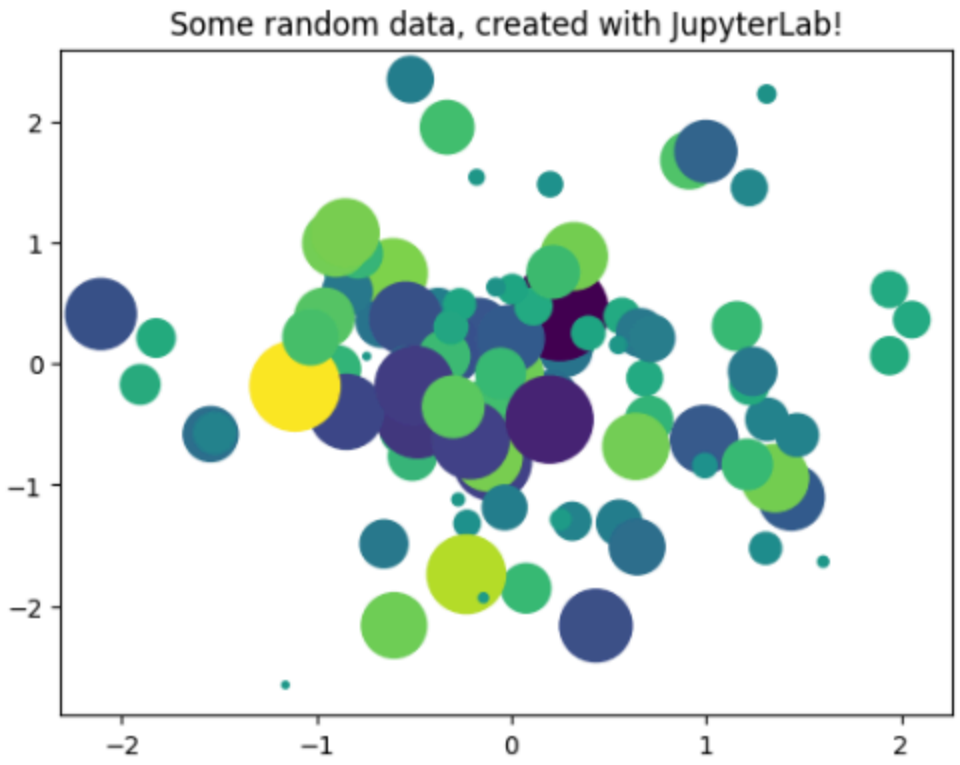
plt.scatter(x, y)
plt.show()
```



```
from matplotlib import pyplot as plt
import numpy as np

#Generate 100 random data points along 3 dimension
x,y, scale = np.random.randn(3,100)
fig, ax = plt.subplots()

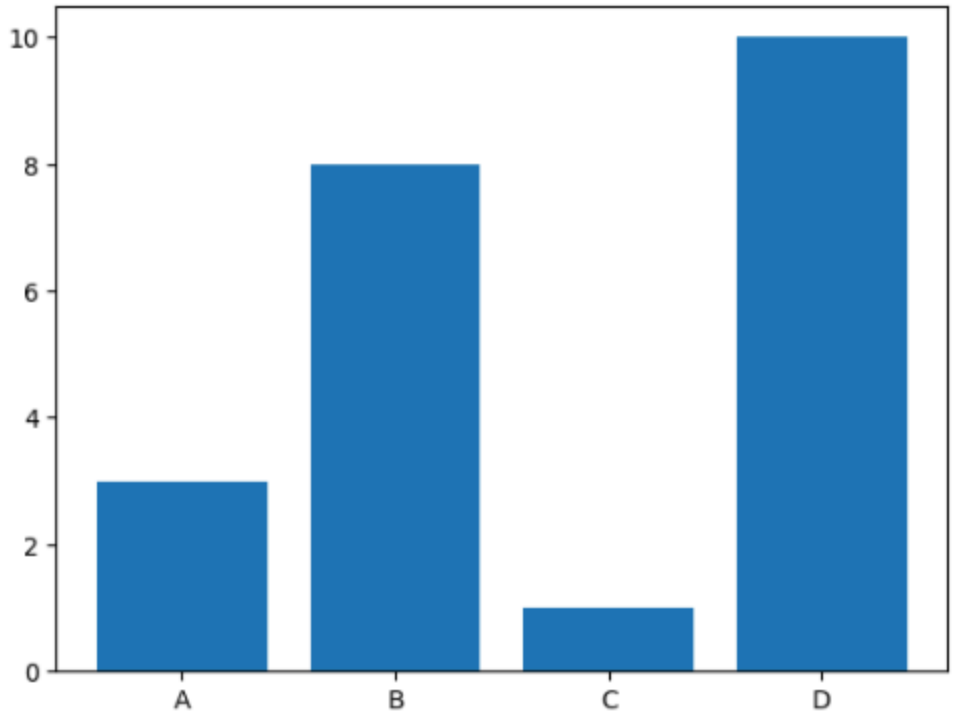
#Map each onto a scatterplot we'll create with Matplotlib
ax.scatter(x=x, y=y, c=scale, s=np.abs(scale)*500)
ax.set(title="Some random data, created with JupyterLab!")
plt.show()
```



```python
import matplotlib.pyplot as plt
import numpy as np

x = np.array(["A", "B", "C", "D"])
y = np.array([3, 8, 1, 10])

plt.bar(x,y)
plt.show()
```



```python
import numpy as np

x = np.random.normal(170, 10, 250)

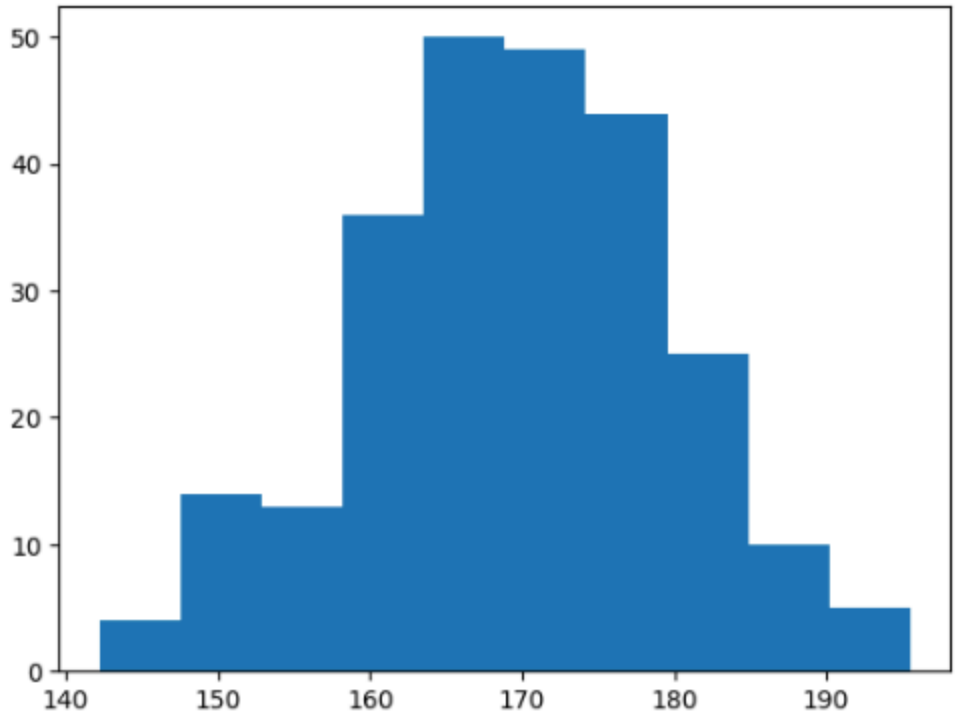
print(x)
```

```
[168.51647187 173.21176238 163.83267273 189.4057519  171.41921303
 171.21351901 159.11881474 177.28628896 157.52496258 177.13246748
 182.22164135 174.67927075 178.77282688 161.73103947 161.98961654
 153.67512709 172.90324278 180.10153081 161.31253846 160.47588323
 177.26759147 171.21349929 147.92291609 182.01771163 175.25340381
 160.13374101 167.67033793 171.49283742 161.7966377  169.51355285
 177.71348113 153.00235433 178.42652507 160.31713747 184.01282695
 187.48068247 157.9879711  182.37547607 169.38258651 164.87736473
 161.5693259  183.49217621 181.85267563 174.57281459 175.68093905
 161.96092349 158.18828777 177.89890963 166.11733934 165.73342263
 169.7548489  161.08141722 176.74528326 162.44670127 156.72007272
 169.969404   175.84410287 173.60988418 166.25244428 189.89709264
 163.77624517 166.50103101 175.4019924  171.6564161  174.35304308
 153.56823824 159.74773105 188.91847652 184.70367356 169.1891274
 162.99597126 158.94822014 183.64881448 185.17720472 164.96379632
 169.37977881 173.70391089 159.99534273 162.82543493 174.15778309
 175.79060043 170.32654152 166.47610745 183.71727714 166.63059697
 163.53383011 169.09424612 151.83425455 164.5709618  178.04146759
 165.26378128 152.36008197 180.50278077 177.57448749 164.8999472
 162.87168276 161.7493862  167.43048291 161.70820085 166.51511888
 154.03514482 166.79749404 172.29538968 164.78581315 172.16272651
 156.32518754 156.13800294 153.10993474 170.89312823 163.97647624
 171.01771143 143.08408967 175.22534934 177.39380619 163.68103278
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 173.52574363 186.30303682 174.60351177 161.32265132 157.35066974
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 168.5789663  180.35835434 187.13332775 158.09305037 156.48567891
 183.60920045 186.90073716 159.40619744 161.72473459 171.96610723
 187.66543143 186.5078725  163.03924734 164.29375936 184.81549676
 165.31968231 175.38327108 154.04797319 179.3720275  175.79339144
 162.46385621 163.44254304 181.20003451 168.69125772 186.90441045
 161.07233421 161.45015578 163.77023028 155.27822801 180.07746976
 164.84412177 174.11420449 169.5795353  159.55382274 161.72890871
 168.67735287 174.95661083 166.53447618 158.04802592 174.94698335
 166.96379653 170.52639432 166.74763467 173.32678764 158.3865986
 173.91120999 178.41815715 160.31302751 167.83388003 163.86336078
 177.82157484 157.68875331 174.63450629 186.88076909 178.16328852
 170.65828304 166.03946896 160.25297915 172.13388948 170.57905592
 178.31043096 164.34595029 164.45660074 177.27552702 161.57634369
 168.12546433 157.93079339 154.73865098 171.39170992 172.62963751
```

```python
import matplotlib.pyplot as plt
import numpy as np

x = np.random.normal(170, 10, 250)

plt.hist(x)
plt.show()
```



```python
import matplotlib.pyplot as plt
import numpy as np

y = np.array([35, 25, 25, 15])

plt.pie(y)
plt.show()
```

